VIT Vidyalankar Institute of Technology Accredited A+ by NAAC	Department of Computer Engineering A.Y-2026-27

Semester	B.E. Semester VII – CMPN
Subject	DevOps
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Assisting Teachers	Prof. Snehal Andhare
Laboratory	

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Grade and Subject Teacher's Signature		

Experiment Number	9
Experiment Title	Luggage Complaint Portal: Selenium Test Design and Local Execution
Objectives (Skill Set / Knowledge Tested / Imparted)	To identify critical user journeys, write Selenium WebDriver tests with assertions and test data, save screenshots on failure and run the suite locally through Maven.
Program	**1. Test plan** **2. Local execution through Maven**

ID	Journey	Steps	Expected result
J1	Valid complaint shows a reference number	Open /complaints/new, fill all fields, submit	Reference in the format LCP-YYYY-NNN NNN
J2	Empty form shows validation errors	Open /complaints/new, submit empty	At least 5 validation errors
J3	Tracking shows the status	Submit a complaint, open /track, enter the reference	Status is SUBMITTED
J4	Reviewer approves a complaint	Submit, log in as reviewer, open complaint, choose APPROVE	Status becomes APPROVED

Test data: Asha Verma, asha.verma@example.com, 9876543210, flight AI202, PNR AB12CD, bag tag 1234567890, type DELAYED. Tool: Selenium WebDriver 4 with JUnit 5, class PortalJourneyTest.

```
mvn -B test -Pselenium -Dtest=PortalJourneyTest
-Dbase.url=http://localhost:8082/luggage-portal
```

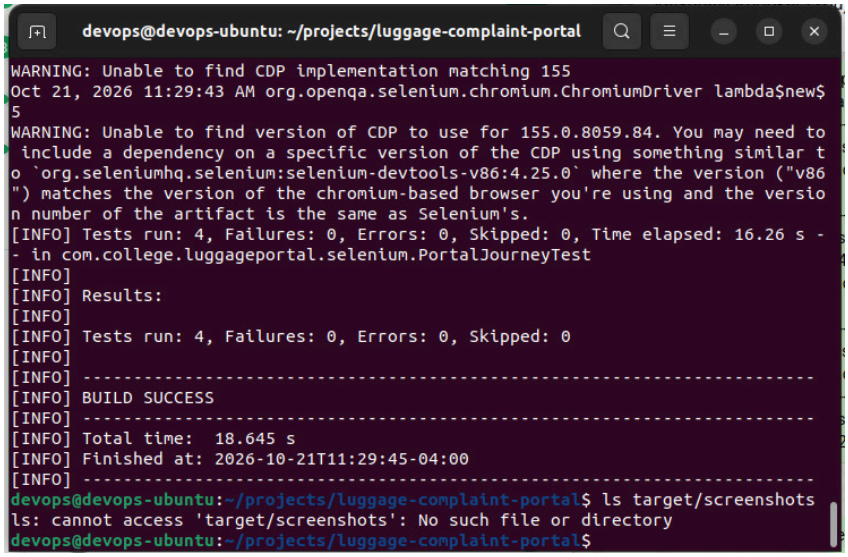


Figure: Tests run: 4, Failures: 0, Errors: 0 and BUILD SUCCESS

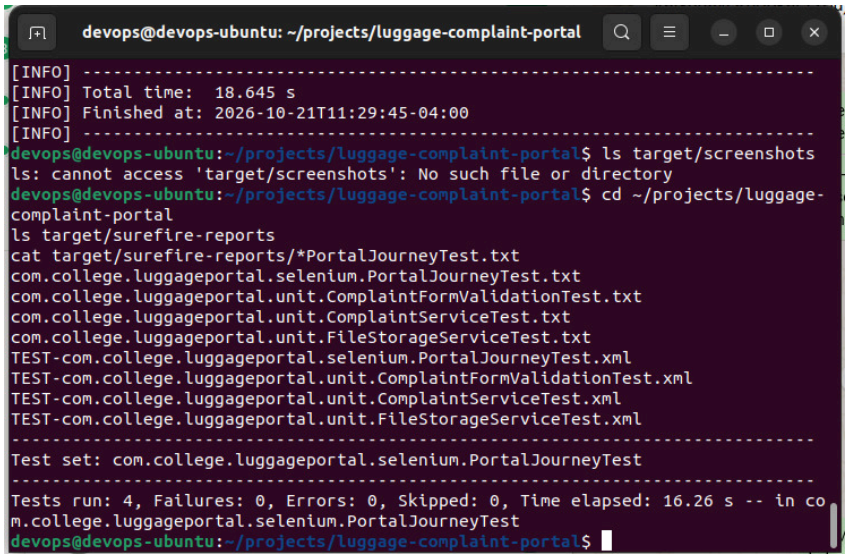


Figure: Surefire reports and the PortalJourneyTest report

3. Failure screenshot mechanism

ScreenshotOnFailure.java is a JUnit 5 extension that saves target/screenshots/<testName>.png when a test fails.

4. Deliberate defect to prove the mechanism

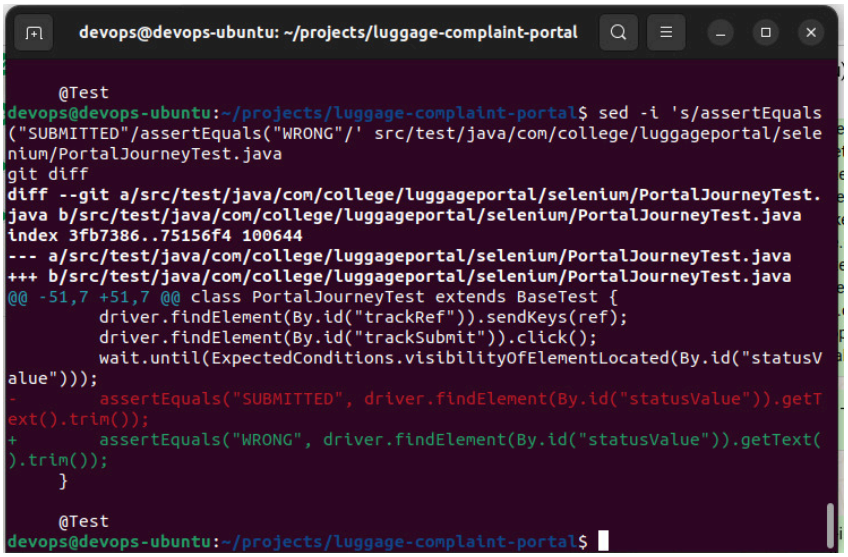


Figure: Expected status changed from SUBMITTED to WRONG in journey J3

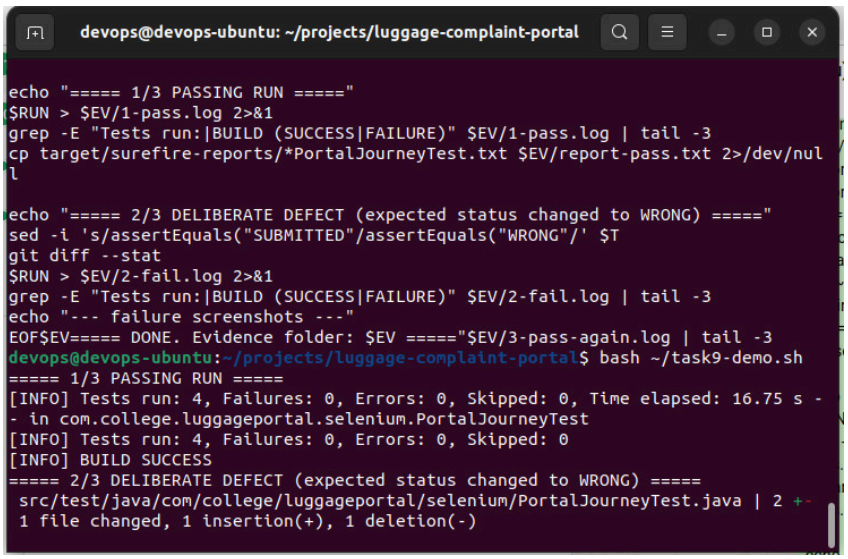


Figure: Demonstration script: passing run, then the deliberate defect

```
===== 1/3 PASSING RUN =====
Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
BUILD SUCCESS
===== 2/3 DELIBERATE DEFECT =====
Tests run: 4, Failures: 1, Errors: 0, Skipped: 0
BUILD FAILURE
--- failure screenshots ---
j3_trackingShowsStatus.png
===== 3/3 FIX AND RERUN =====
Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
BUILD SUCCESS
```

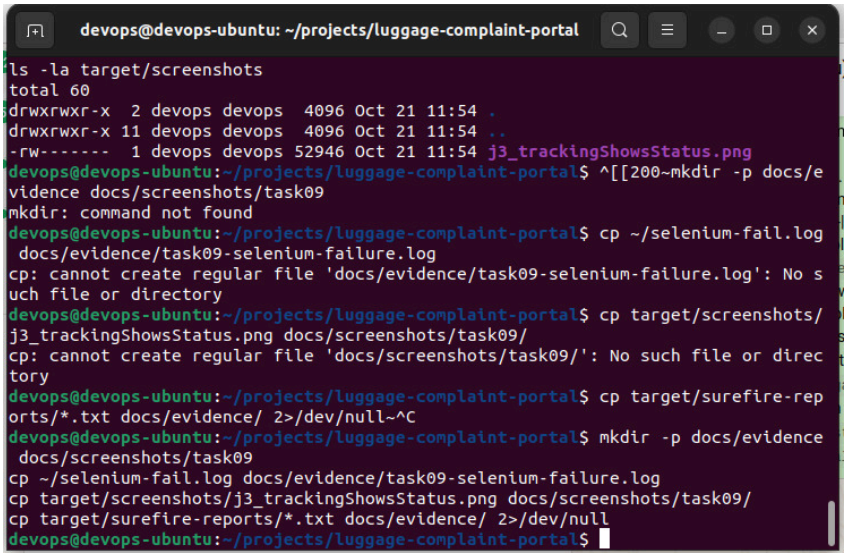


Figure: Saved failure screenshot in target/screenshots

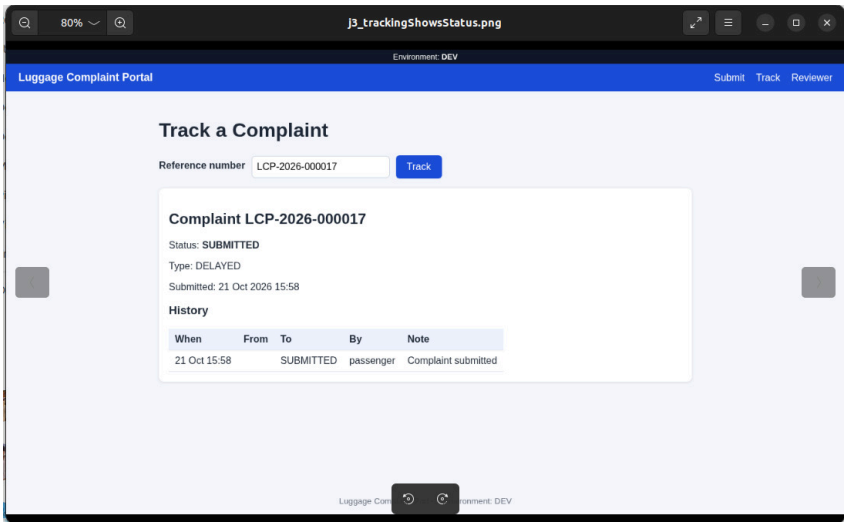


Figure: j3_trackingShowsStatus.png captured automatically at the failure

Conclusion	Four critical journeys were automated with Selenium and run through Maven. A screenshot is saved automatically when a test fails, and the suite passed again after the defect was removed.